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[shiyiping@panda NTT8Kara8_final]\$ python Poly_NTT8_Kara8.py

Parameter Definition

N = 65536

K = 32

q = 998244353

phi = 629671588

phi_inv = 283043518

psi = 24514907

psi_inv = 3707709

N_inv = 998229121

inv_2 = 499122177

0. Initialization

('A[0:16]: ', [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]) ('B[0:16]: ', [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15])

1. Negacyclic convolution (Preprocess)

('Nega_conv Time: ', 3369.5423288345337)

2. NTT (8 stages)

('NTT Time: ', 5925.766027927399)

3. Karatsuba (8 stages)

('Karatsuba Time: ', 2.1045401096343994)

4. Element-wise Multiplication

('Element-wise Multiplication Time: ', 0.5550050735473633)

5. Inv-Karatsuba (8 stages)

('Inv_Karatsuba Time: ', 4.856817007064819)

6. Inv-NTT (8 stages)

('INTT Time: ', 5587.164417028427)

7. Negacyclic convolution (Postprocess)

('C[0:16]: ', [995506068L, 844609430L, 693778330L, 543012770L, 392312752L, 241678278L, 91109350L, 938850323L, 788412493L, 638040215L, 487733491L,

337492323L, 187316713L, 37206663L, 885406528L, 735427604L])

('Inv_Nega_conv Time: ', 1402.7926428318024)

Verification

('Base[0:16]: ', [995506068, 844609430, 693778330, 543012770, 392312752, 241678278, 91109350, 938850323, 788412493, 638040215, 487733491, 337492323, 187316713, 37206663, 885406528, 735427604])

C == Base --> Verification PASS! 😊 😊 😊 😊 😊

Total Time consumption

('Execution Time: ', 17318.86577105522)